

Problem 25.9

Payments will be made to you at a continuous rate of \$100 per year for the next five years. The effective rate of interest over this period is $i = 0.06$. Calculate the present value of this payment stream.

Solution

The payments are made continuously at a rate of 100 per year for 5 years. Thus, the present value is

$$100\bar{a}_{\overline{5}|}.$$

Since the annual effective interest rate is

$$i = 0.06,$$

the corresponding force of interest is

$$\delta = \ln(1 + i) = \ln(1.06).$$

For a continuous annuity,

$$\bar{a}_{\overline{5}|} = \frac{1 - e^{-5\delta}}{\delta}.$$

Because $e^{-5\delta} = (1.06)^{-5}$,

$$\bar{a}_{\overline{5}|} = \frac{1 - (1.06)^{-5}}{\ln(1.06)}.$$

Therefore,

$$PV = 100 \left(\frac{1 - (1.06)^{-5}}{\ln(1.06)} \right).$$

Thus,

$$PV = 433.7507519.$$

So the present value is

$$\boxed{433.75}.$$